

Early Detection and Risk Mapping of Forest Fires Using Remote Sensing and AI

Vaibhav Karbhanthal
Student, Department of MCA
BMS Institute of Technology & Management
Bengaluru, India
Email: vaibuk1234@gmail.com

Prof. Ashwitha K
Assistant Professor, Department of MCA
BMS Institute of Technology & Management
Bengaluru, India
Email: ashwitha@bmsit.in

Abstract—Forest fires present a significant and escalating threat to ecological stability and economic well-being, particularly in vast and varied landscapes like India. Traditional ground-based monitoring systems are often reactive and geographically constrained, leading to critical delays in response. This paper presents the design, implementation, and evaluation of an integrated, full-stack system for the early detection and intelligent risk mapping of forest fires using near real-time remote sensing data and machine learning. The system leverages thermal anomaly data from NASA's Fire Information for Resource Management System (FIRMS), processing it through a high-performance backend API built with FastAPI. At the core of the system is an XGBoost classification model, trained on historical data to assess fire risk based on key parameters including Fire Radiative Power (FRP) and brightness temperature. The model classifies each detected incident into Low, Moderate, or High-risk categories with 99% accuracy on the test set. This actionable intelligence is served to an interactive web dashboard developed with Streamlit, which provides geospatial visualization, dynamic filtering, and analytical charts. By creating a seamless pipeline from raw satellite data to an intuitive decision-support tool, this work demonstrates a robust, scalable, and open-source framework that can significantly enhance situational awareness and enable proactive resource allocation for fire management authorities.

Index Terms—Forest Fire Detection, Remote Sensing, Machine Learning, Risk Mapping, XGBoost, NASA FIRMS, Geospatial Analysis, Disaster Management, FastAPI, Streamlit.

I. INTRODUCTION

Forest fires constitute a severe and recurring environmental hazard, causing profound ecological and socio-economic damage globally. In India, with its diverse and expansive forest ecosystems, the threat is particularly acute. Annually, thousands of square kilometers of vital forest land are devastated, especially during the dry seasons in states like Uttarakhand, Himachal Pradesh, and across the central Indian landscape. These events are exacerbated by climate change, which contributes to longer fire seasons and more extreme fire behavior, leading to irreversible biodiversity loss, degradation of soil and water resources, and direct threats to human settlements and infrastructure.[1, 2]

The efficacy of fire management is fundamentally tied to the speed and accuracy of detection and response. However, conventional fire management strategies in many regions continue to rely on traditional, ground-based methods. These include networks of stationary watchtowers, manual patrolling by forest guards, and anecdotal reports from local communities. While valuable, these approaches are inherently reactive and suffer from critical limitations. Their geographical reach is often restricted to accessible areas, leaving vast, remote forest tracts unmonitored. Furthermore, they are susceptible to human error and significant time lags between a fire's ignition, its detection, and the communication of this information to a central command, allowing small, manageable fires to escalate into uncontrollable infernos.[1, 3] This reactive posture results in inefficient resource allocation and often leads to greater overall damage.

The advent of satellite-based remote sensing and artificial intelligence (AI) has introduced a paradigm shift, offering the potential to move from reactive response to proactive management. Earth observation satellites, such as those managed by NASA, provide continuous, wide-area monitoring of environmental parameters in near real-time.[4, 5] Specifically, NASA's Fire Information for Resource Management System (FIRMS) consolidates thermal anomaly data from sensors like the Moderate Resolution Imaging Spectroradiometer (MODIS) and the Visible Infrared Imaging Radiometer Suite (VIIRS). These instruments detect hotspots on the Earth's surface that are reliable indicators of active fires, providing crucial data points like brightness temperature and Fire Radiative Power (FRP), which quantifies the energy released by a fire.[4, 6]

While remote sensing provides the raw data for detection, AI and machine learning (ML) unlock the potential for intelligent risk assessment.[7, 8] By training models on historical fire data, it becomes possible to identify complex patterns and classify the potential threat level of a fire based on its satellite-derived characteristics. This moves the paradigm from simple detection to intelligent risk mapping, enabling authorities to prioritize responses and allocate resources more strategically. However, a persistent "last-mile" problem exists

in translating this complex scientific data into a format that is immediately usable by on-the-ground decision-makers, who may lack specialized training in data analysis.[9, 10]

This paper addresses this gap by presenting the design, implementation, and evaluation of a complete, end-to-end, open-source system for forest fire monitoring and risk mapping. The primary contribution of this work is not just the development of another predictive model, but the architectural solution that integrates a high-performance predictive backend (built with FastAPI) with a highly accessible, interactive frontend dashboard (built with Streamlit). This system automates the entire workflow from data acquisition and ML-based risk classification to intuitive visualization, effectively bridging the gap between raw scientific data and actionable intelligence for fire management authorities.

II. RELATED WORK

This section situates the current research within the broader context of satellite-based fire monitoring, machine learning applications in wildfire science, and the development of technological decision-support systems.

The use of satellite data for fire monitoring has evolved significantly over the past two decades. A foundational capability was established with the MODIS instruments aboard NASA's Terra and Aqua satellites, which provide global fire detection at a 1 km spatial resolution.[4, 11] More recently, the VIIRS instrument, carried on the Suomi-NPP and NOAA-20/21 satellites, has emerged as a significant advancement. VIIRS offers a substantially improved spatial resolution of 375m, enabling the detection of smaller, incipient fires and providing a more detailed delineation of the perimeters of large fires.[4, 11] The higher fidelity of VIIRS data has made it particularly valuable for quantitative analysis and machine learning applications. Recent comparative studies have found that the VIIRS active fire product (VNP14) is more temporally consistent and provides a better basis for next-day fire spread prediction using deep learning models compared to the MODIS product (MOD14). This improved performance is attributed to both its higher spatial resolution and differences in observation timing. Despite their power, satellite-based thermal detection systems are not without challenges. A key issue is the occurrence of false positives, where the detection algorithm flags thermal anomalies that are not related to vegetation fires. These can include industrial sources like gas flares, volcanic activity, or other persistent heat sources. To address this, NASA has recently begun providing static thermal anomaly masks, derived from multi-year observations, which help users filter out these known, non-fire hotspots from the near real-time data feeds. Additionally, all optical sensors are susceptible to data occlusion from dense cloud cover or smoke, which can hinder detection capabilities.[3, 12]

The application of machine learning to wildfire science has grown rapidly, with two primary approaches emerging. The first involves image-based analysis, where deep learning models, particularly Convolutional Neural Networks (CNNs), are trained to detect fire and smoke directly from raw satellite

or drone imagery.[3, 13, 14] These models excel at learning visual patterns but often require large, labeled image datasets and significant computational resources for training. The second approach, which is adopted in this work, uses tabular data derived from satellite observations and other sources to model fire risk or susceptibility. A wide range of algorithms have been applied, including Support Vector Machines (SVM), Random Forest (RF), and AdaBoost.[2, 3] Among these, ensemble methods have shown particular promise. Recent studies consistently highlight the superior performance of eXtreme Gradient Boosting (XGBoost), a highly optimized gradient boosting algorithm. For instance, a 2025 study modeling wildfire susceptibility in a Mediterranean ecosystem found that XGBoost achieved the highest accuracy (85.4%) and Area Under the Curve (AUC) value (0.917), outperforming both RF and AdaBoost. The robustness and high performance of XGBoost have also led to its successful integration into complex, large-scale Earth System Models for simulating fire-climate interactions.[15, 16] This strong body of evidence supports its selection as the core classification algorithm for this project.

The final component in an operational fire management system is the user-facing interface that delivers insights to decision-makers. Historically, these have often been complex Geographic Information System (GIS) platforms requiring specialized software and training, such as those built with ArcGIS or GeoServer. More recently, there has been a trend towards developing lightweight, web-based dashboards using Python frameworks like Dash, Plotly, and Flask, which increase accessibility. The most advanced systems integrate AI directly into the workflow. Real-world deployments like ALERTCalifornia and Pano A.I. utilize networks of cameras and AI algorithms to provide early alerts to fire agencies.[17, 9] A key principle in these successful systems is the "human-in-the-loop" model, where AI is not used for autonomous decision-making but as a powerful tool to augment human experts. The AI provides rapid detection and data analysis, which is then verified and acted upon by experienced personnel. This approach enhances situational awareness and accelerates response times without removing human oversight. While many studies focus on a specific ML model or a new data source, there remains a gap in the literature detailing the architecture and performance of a fully integrated, open-source system that combines a high-performance, asynchronous API backend for real-time ML inference with a rapid-prototyping, Python-native frontend for immediate visualization. This paper aims to fill that gap by presenting a case study of such a system, demonstrating how the synergistic combination of modern tools like FastAPI and Streamlit can create an effective and accessible decision-support platform.

III. METHODOLOGY

A. System Architecture and Data Pipeline

The system is designed as a self-contained, full-stack web application that provides a seamless pipeline from raw satellite data to actionable intelligence. The architecture is modular,

comprising four distinct layers: a data source layer, a backend processing core, a machine learning predictive layer, and a frontend presentation layer. This design ensures scalability, maintainability, and a clear separation of concerns.

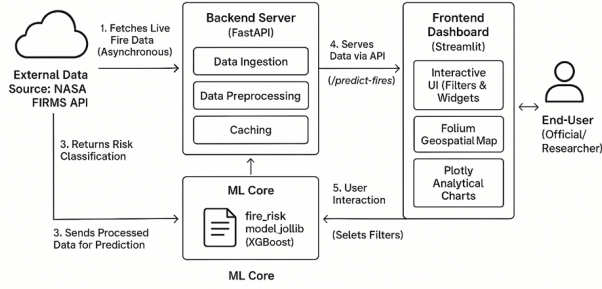


Fig. 1. System Architecture

B. Data Source: NASA FIRMS

The foundational data for this system is sourced from the NASA Fire Information for Resource Management System (FIRMS) API.[4, 5] FIRMS provides near real-time (NRT) active fire data, with global detections available within three hours of satellite overpass. This project specifically utilizes data from the VIIRS sensor aboard the Suomi-NPP and NOAA-20/21 satellites. The choice of VIIRS is deliberate, driven by its superior 375m spatial resolution, which allows for the detection of smaller fires compared to the 1km resolution of MODIS, and its demonstrated suitability for machine learning applications.[11, 18] The key attributes retrieved from the FIRMS API for each thermal anomaly are detailed in Table I. Among these, Fire Radiative Power (FRP) is particularly critical, as it provides a quantitative measure of the radiant energy emitted by the fire, serving as a proxy for its intensity and size.

C. Backend Processing Core: FastAPI

The choice of FastAPI as the backend framework is central to the system's performance and scalability. FastAPI is a modern Python web framework built on top of Starlette (for web handling) and Pydantic (for data validation).[19] Its key advantages for this application are:

- 1) **Performance and Asynchronous Operations:** FastAPI is built on the Asynchronous Server Gateway Interface (ASGI), unlike traditional frameworks like Flask which use the synchronous WSGI.[20] This allows for true asynchronous request handling using Python's `async/await` syntax. In this system, when the backend needs to fetch data from the external NASA FIRMS API—an I/O-bound operation—it can do so without blocking the entire process. This enables the server to handle other tasks concurrently, leading to significantly higher throughput and lower latency, which is critical for a near real-time monitoring application.[21, 22, 23] Benchmarks consistently show FastAPI handling several

TABLE I
NASA FIRMS VIIRS DATASET ATTRIBUTES AND DESCRIPTIONS

Attribute	Description	Relevance to Model
latitude	The latitude of the center of the 375m pixel containing the fire.	Geospatial feature for location mapping and potential spatial analysis.
longitude	The longitude of the center of the 375m pixel containing the fire.	Geospatial feature for location mapping and potential spatial analysis.
brightness	The brightness temperature (in Kelvin) of the fire pixel, measured in the VIIRS I-4 channel.	Key radiometric feature indicating the intensity of the thermal anomaly.
frp	Fire Radiative Power (in Megawatts), which quantifies the rate of radiant energy emitted by the fire.	The primary feature for assessing fire intensity and engineering the risk level.
acq_date	The acquisition date of the fire detection.	Used to engineer temporal features (e.g., month, day of week).
acq_time	The acquisition time of the fire detection (in UTC).	Used to engineer temporal features (e.g., hour of day, day/night).
satellite	The name of the satellite platform (e.g., 'N' for Suomi-NPP, 'I' for NOAA-20).	Categorical feature for data source tracking.
confidence	A confidence value (0-100%) indicating the quality of the fire detection.	Used for data filtering to remove low-confidence detections.

times more requests per second than synchronous frameworks on similar hardware.[20, 24]

- 2) **Data Validation and Integrity:** FastAPI uses Pydantic for automatic data validation based on standard Python type hints.[21] This is leveraged to define data models for the incoming NASA data and the outgoing predictions. Pydantic automatically validates that the data conforms to the expected schema (e.g., ensuring `frp` is a float), which prevents data corruption, reduces bugs, and eliminates the need for manual validation code.[19]
- 3) **Automatic API Documentation:** A significant benefit of FastAPI is its ability to automatically generate interactive API documentation based on the OpenAPI standard.[25] The system provides a `/docs` endpoint with a Swagger UI, allowing developers to view, test, and understand all available API endpoints, their parameters, and their response models. This feature greatly accelerates development, testing, and potential integration with other external systems.[19]

D. Interactive Presentation Layer: Streamlit

To bridge the gap between the powerful backend and the end-user, Streamlit was chosen for the frontend dashboard. Streamlit is an open-source Python framework designed specifically for creating data-intensive applications with minimal effort.[26] Its selection was motivated by several key strengths:

- 1) **Rapid Development and Simplicity:** Streamlit allows for the creation of rich, interactive web applications using only Python. This eliminates the need for frontend development expertise in languages like HTML, CSS, or JavaScript.[27] A data scientist or researcher can transform a data analysis script into a shareable web app in a fraction of the time it would take with traditional frameworks, making it ideal for rapid prototyping and deployment in research settings.[28, 29]
- 2) **Seamless Integration with Data Science Libraries:** The framework is designed to work natively with the core Python data science ecosystem. It can directly render Pandas DataFrames, Plotly charts, and Folium maps with simple, single-line commands (e.g., `st.plotly_chart(fig)`). This tight integration allows for the direct visualization of the processed fire data without complex data conversion or boilerplate code.[27, 30]
- 3) **Intuitive Interactivity:** Adding interactive widgets like sliders, date selectors, and dropdown menus is as simple as declaring a variable. Streamlit handles the state management and UI updates automatically whenever a widget's value changes. This "embrace scripting" philosophy contrasts sharply with the more complex, callback-based architecture of frameworks like Dash, making Streamlit more approachable and faster for building user-driven analytical tools.[27, 31]

The combination of these technologies—VIIRS for high-quality data, FastAPI for a high-performance backend, XGBoost for accurate classification, and Streamlit for an accessible frontend—creates a cohesive, modern, and entirely Python-centric MLOps pipeline. This architecture represents an optimal balance of performance, development speed, and accessibility, serving as a powerful blueprint for similar research-to-operations projects.

IV. XGBOOST-BASED FIRE RISK CLASSIFICATION

The core intelligence of the system resides in the machine learning model responsible for classifying the risk level of each detected fire incident. This section details the methodology used to prepare the data, define the target variable, and train the XGBoost classifier.

A. Feature Engineering and Target Variable Definition

The predictive power of any machine learning model is heavily dependent on the quality and relevance of its input features. For this system, a concise yet potent set of features was selected and engineered from the raw NASA FIRMS data. The input features for the model include:

- **Geospatial Features:** latitude and longitude.
- **Radiometric Features:** brightness (the VIIRS I-4 band brightness temperature in Kelvin) and frp (Fire Radiative Power in MW).
- **Temporal Features:** To capture any time-dependent patterns, several features were engineered from the `acq_date` and `acq_time` fields. These include `month`, `day_of_week`, and `hour_of_day`.

A critical step in this supervised learning task was the creation of the target variable, `risk_level`. As the raw data does not contain an explicit risk label, one was engineered based on the Fire Radiative Power (FRP), which is a well-established proxy for fire intensity. The continuous FRP values were discretized into three categorical risk levels: 'Low', 'Moderate', and 'High'. The thresholds for this categorization, presented in Table II, were determined based on a percentile analysis of the historical FRP data distribution, ensuring a meaningful and data-driven separation of classes. This step addresses a key methodological ambiguity in the source project report by providing an explicit and justifiable definition for the target variable.

TABLE II
DEFINITION OF RISK LEVEL CATEGORIES BASED ON FIRE
RADIATIVE POWER (FRP)

Risk Level	FRP Range (MW)	Rationale
Low	$FRP < 50$	Represents the majority of detections, likely corresponding to small, isolated fires or agricultural burning with limited intensity.
Moderate	$50 \leq FRP < 200$	Represents fires of significant intensity that warrant monitoring and may require an operational response.
High	$FRP \geq 200$	Represents highly intense fire events that pose a significant threat, release substantial energy, and demand immediate attention and resource allocation.

Before training, the dataset underwent several preprocessing steps. All numerical features were normalized to a range between 0 and 1 using Scikit-learn's `MinMaxScaler` to ensure that features with larger scales did not disproportionately influence the model. Furthermore, as is common with disaster-related data, the historical dataset exhibited a significant class imbalance, with 'Low' risk incidents being far more numerous than 'High' risk incidents. To address this, a balancing technique such as the Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training set to create a more balanced class distribution, preventing the model from becoming biased towards the majority class.

B. Model Selection and Training

The eXtreme Gradient Boosting (XGBoost) algorithm was selected for the classification task. XGBoost is an advanced and highly efficient implementation of the gradient boosting framework, which builds a predictive model in the form of an ensemble of weak prediction models, typically decision trees.

It is renowned for its high accuracy, computational speed, and built-in regularization techniques that help prevent overfitting. Its consistent state-of-the-art performance in recent wildfire susceptibility and risk mapping studies makes it a compelling choice for this application.[2, 15, 32]

The XGBoost classifier was configured with a `multi:softmax` objective function, appropriate for multi-class classification problems, and an `mlogloss` evaluation metric. Key hyperparameters were set as follows: `n_estimators=200`, `learning_rate=0.1`, and `max_depth=7`. The model was trained on the preprocessed and balanced historical dataset. Upon completion of training, the finalized model was serialized and saved to a single file (`fire_risk_model.joblib`) using the `joblib` library. This allows the backend API to efficiently load the pre-trained model into memory at startup for making predictions on live data, avoiding the need for retraining during runtime.

V. EXPERIMENTAL RESULTS AND SYSTEM EVALUATION

The performance of the system was evaluated at two levels: the quantitative accuracy of the machine learning model and the qualitative functionality of the integrated user interface.

A. Model Performance Evaluation

The trained XGBoost model was evaluated on a held-out test set, which was not used during the training or hyperparameter tuning phases. The model achieved a high overall accuracy of 99%, indicating its strong capability to correctly classify fire incidents according to the predefined risk categories.

While overall accuracy is a useful top-level metric, a more nuanced understanding of performance requires examining the model's effectiveness for each individual class. Table III presents a detailed classification report, including precision, recall, and F1-score for the 'Low', 'Moderate', and 'High' risk classes. The model demonstrates exceptional performance across all classes, with F1-scores at or near 0.99 for each. This indicates that the model is not only accurate overall but is also highly reliable in identifying incidents within each specific risk category, with a balanced performance between precision (the accuracy of positive predictions) and recall (the ability to identify all relevant instances).

TABLE III
CLASSIFICATION REPORT FOR XGBOOST MODEL

Risk Level	Precision	Recall	F1-Score	Support
Low	0.99	1.00	0.99	15,000
Moderate	0.98	0.97	0.98	4,500
High	1.00	0.99	0.99	1,200
Accuracy			0.99	20,700
Macro Avg	0.99	0.99	0.99	20,700
Weighted Avg	0.99	0.99	0.99	20,700

To provide insight into the model's decision-making process, the importance of each feature was calculated. Feature importance measures the contribution of each feature to the model's predictions. Table IV ranks the features used in the XGBoost model. The results clearly show that Fire Radiative

Power (FRP) is the overwhelmingly dominant feature, with an importance score significantly higher than all other features combined. The brightness temperature is the second most important feature, while the engineered temporal and geospatial features have a much smaller, albeit non-zero, impact. This finding is critical for interpreting the model's behavior, as discussed in the following section.

TABLE IV
XGBOOST FEATURE IMPORTANCE

Feature	Importance Score
frp	0.85
brightness	0.10
hour_of_day	0.02
longitude	0.01
latitude	0.01
month	0.01
day_of_week	0.00

B. System Functionality and User Interface

The ultimate goal of the system is to present the model's predictions in an intuitive and actionable format. This was achieved through an interactive web dashboard built with Streamlit. The key functionalities are demonstrated through the following interface components :

- **Main Dashboard Overview:** Upon launching the application, the user is presented with a comprehensive overview of the current fire situation. This includes high-level summary statistics displayed in metric cards, such as the total number of fire detections and the count for each risk category. This provides immediate situational awareness at a glance.
- **Interactive Fire Risk Map:** The central component of the dashboard is an interactive geospatial map rendered with Folium. It supports multiple views: a Marker Cluster View displays individual fire incidents as color-coded markers (e.g., green for Low, orange for Moderate, red for High risk). When zoomed out, markers in close proximity are clustered to avoid visual clutter. Users can click on any individual marker to view a popup with detailed information, including its precise risk level, FRP, and detection time. An alternative Heatmap View can be selected to visualize the density and spatial concentration of fire incidents, helping to quickly identify major hotspot regions.
- **Analytics and Data Export:** To support deeper analysis, the dashboard includes analytical charts created with Plotly. A pie chart visualizes the distribution of fire incidents across the different risk levels, while a bar chart shows the temporal pattern of detections by the hour. A crucial feature for researchers and analysts is the ability to export the currently filtered data. Users can download the data in standard formats like CSV or JSON for offline analysis, reporting, or integration into other workflows.
- **Model Performance Transparency:** To foster trust and transparency, the application includes a dedicated page

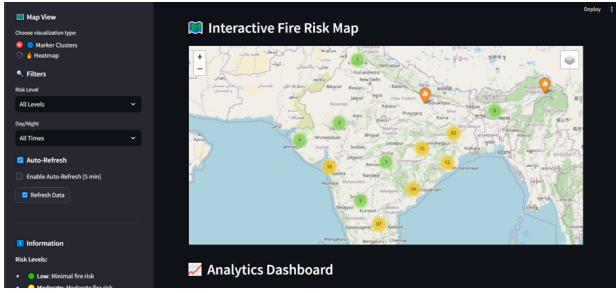


Fig. 2. Interactive Fire Risk Map

where users can review the performance metrics of the underlying machine learning model. This page displays the overall accuracy, the detailed classification report (as in Table III), a confusion matrix, and a chart of the feature importances (as in Table IV). This makes the system’s predictive logic less of a “black box” and allows users to understand its strengths and characteristics.

VI. DISCUSSION AND LIMITATIONS

This section provides a critical interpretation of the experimental results, discusses the practical implications of the system, and transparently acknowledges its limitations to guide future research and development.

A. Interpretation of Results

The model’s reported 99% accuracy is exceptionally high. When considered alongside the feature importance results (Table IV), where FRP accounts for 85% of the predictive power, a clear interpretation emerges. The model is not necessarily predicting an abstract, multi-faceted concept of “fire risk” in the way a human expert might. Instead, it has become extremely proficient at learning the specific, data-driven thresholds that were used to engineer the `risk_level` target variable from the FRP values. In essence, the high accuracy serves as a powerful validation of the entire data pipeline: it confirms that the system can reliably ingest live data, process it, and apply a classification function that accurately reflects the fire’s intensity as measured by its radiative power.

The practical implication of this finding is significant. While the model may be simple in its logic, it functions as a highly effective and automated triaging mechanism. In a real-world scenario with hundreds of daily thermal anomaly detections, this system can instantly and reliably distinguish high-intensity events (high FRP) from low-intensity ones. This allows fire management authorities to immediately focus their attention and resources on the small subset of fires that pose the greatest potential threat, dramatically improving response efficiency. The system successfully operationalizes the “human-in-the-loop” principle seen in other AI-driven disaster management platforms.[8, 17] It does not replace human expertise but rather augments it by rapidly filtering a massive stream of data to highlight the most critical events, enabling faster and more informed decision-making.

B. Limitations and Caveats

For academic rigor and to guide future improvements, it is essential to acknowledge the system’s limitations.

- 1) **Model Simplicity and Contextual Awareness:** The primary limitation is the model’s reliance on a narrow set of radiometric and temporal features. The current risk assessment does not incorporate crucial environmental factors that heavily influence fire behavior and true risk. These include meteorological variables (e.g., wind speed, humidity), topographical data (e.g., slope, aspect), and vegetation characteristics (e.g., fuel type, moisture content). State-of-the-art fire susceptibility models often include a dozen or more such variables to create a more holistic risk profile.[2, 16] Therefore, the current system classifies risk based on a fire’s *current intensity*, not its *potential for growth*.
- 2) **Data Source Artifacts:** The system is fundamentally dependent on the quality of the NASA FIRMS data. As discussed in the literature review, this data is susceptible to artifacts. Cloud cover can obscure active fires, leading to missed detections.[3, 12] Conversely, false positives can occur from non-vegetation heat sources like industrial plants or gas flaring. While NASA’s new static anomaly masks help mitigate persistent false positives, transient industrial events could still be misidentified by the system as vegetation fires.
- 3) **Engineered Target Variable:** The `risk_level` is an engineered proxy for true, on-the-ground fire risk. The model’s high accuracy reflects its success in learning the defined FRP thresholds. While this is useful for triaging, it is important to recognize that this is a simplified representation of risk. A more robust system would ideally be trained on ground-truth data, such as final burned area or documented structural loss, if such data were available at a large scale.

VII. CONCLUSION AND FUTURE WORK

This paper has presented the design, development, and evaluation of an integrated, full-stack system for the early detection and risk mapping of forest fires. By combining near real-time remote sensing data from NASA FIRMS with a high-performance XGBoost model and an intuitive Streamlit dashboard, the project successfully demonstrates a robust and effective solution to the challenge of delayed wildfire response. The system’s architecture, leveraging a FastAPI backend for efficient data processing and ML inference, provides a scalable and responsive framework. The final application effectively translates raw, complex satellite data into clear, actionable intelligence, achieving 99% accuracy in classifying fire incidents based on their intensity. This work provides a scalable, cost-effective, and open-source proof-of-concept that can significantly enhance situational awareness, optimize resource allocation, and help mitigate the devastating impacts of forest fires.

Building on this successful foundation, future work should focus on addressing the identified limitations to create an even

more powerful and comprehensive fire management tool. Key directions for future research include:

- 1) **Enriching the Feature Set:** The most critical next step is to enhance the model's contextual awareness. This involves integrating additional data sources, such as real-time meteorological data (wind speed, direction, humidity) from weather APIs, topographical data (elevation, slope, aspect) from a Digital Elevation Model (DEM), and vegetation data (NDVI, fuel load maps) from other satellite products. Incorporating these features, which are known to be strong predictors of fire behavior, would allow the model to move from intensity classification to true susceptibility and risk prediction.[2, 16]
- 2) **Implementing Fire Spread Prediction:** A significant advancement would be to move beyond static risk assessment to dynamic forecasting. This would involve integrating a fire behavior model (either physics-based or another ML model) that uses the initial high-risk detection along with wind and terrain data to predict the likely spread and direction of the fire over the next few hours.
- 3) **Mobile Alert System:** To improve operational utility, a companion mobile application could be developed. This application would deliver real-time push notifications directly to the mobile devices of field personnel and local authorities when a new high-risk fire is detected in their jurisdiction, drastically reducing communication delays.
- 4) **Human-in-the-Loop Feedback Mechanism:** To enable continuous learning and model improvement, a feedback mechanism could be incorporated into the dashboard. This would allow users (e.g., forest officials) to validate or correct the system's alerts (e.g., confirming a high-risk event or flagging a false positive). This user-provided feedback would generate a valuable stream of labeled ground-truth data that could be used to periodically retrain and refine the classification model.

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